

□ □ □ □ □ □ □ □ □ □ □ □

□ □ □ V1.0
□ □ □ 2026 7 13
□ □ □ □ M4
□ □ □ □

1.

□ □ □ □	□ □ □ □
M4	FreeRTOS
	API MQTT

□ □ □ □ □ GPS GPS

2.

2.1

□□	□□□□	□□
FR-TER-001	GStreamer	
FR-TER-002	imx_rpmsg_tty	A M4
FR-TER-003		
FR-TER-004	/dev/ttyUSB2 IMEI	
FR-TER-005	3	
FR-TER-006	DEFAULT M4 INITIALIZED	
FR-TER-007	RPMsg MQTT	

2.2 A M4

□□	□□□□	□□
FR-COM-001	/dev/ttyRPMMSG30 M4	1
FR-COM-002	0x11 0x45 0x14 JSON	
FR-COM-003	ctrl_byte sensor_acc_x sensor_acc_y sensor_acc_z	100
FR-COM-004	5	
FR-COM-005	M4 INITIALIZED PUSHING RECORDING DEFAULT	3
FR-COM-006		4096 512

2.3

□□	□□	□□□□	□□□□□
FR-CTL-001	SOS		SOS M4 PUSHING
FR-CTL-002			pushing M4 PUSHING
FR-CTL-003			recording M4 RECORDING
FR-CTL-004			DEFAULT

□□	□□	□□□□	□□□□□
FR-CTL-005		10%	
FR-CTL-006		10%	
FR-CTL-007		0xFF	
FR-CTL-008		DEFAULT SOS	

2.4

□□	□□□□	□□□□□
FR-DAT-001	MQTT	5 20
FR-DAT-002	1	QoS 0
FR-DAT-003	battery helmetStatus acc_x acc_y acc_z latitude longitude	JSON
FR-DAT-004		
FR-DAT-005	helmetStatus	
FR-DAT-006	helmet/device/114 514	
FR-DAT-007		70 30.593 114.305

2.5

□□	□□□□	□□□□
FR-MED-001	RTSP	640×480 30 H.264 bitrate 4000
FR-MED-002	RTSP	ALSA AAC bitrate 32 000 4.0
FR-MED-003		RTSP
FR-MED-004		10 300
FR-MED-005		NULL
FR-MED-006	0 640×480 JPG	/mnt/mmc/result/ image
FR-MED-007	0 YUYV 30 AVI	/mnt/mmc/result/ video
FR-MED-008		/home/root/temp/result

2.6

□□	□□□□
FR-AUD-001	
FR-AUD-002	
FR-AUD-003	
FR-AUD-004	pactl 10%

3. M4

□□	□□□□	□□
FR-M4-001	RDC	
FR-M4-002	I2C3 16 MHz	3
FR-M4-003	cJSON FreeRTOS	pvPortMalloc vPortFree
FR-M4-004	FreeRTOS	FREERTOS_Init vTaskStartScheduler
FR-M4-005		

4.

4.1

□□	□□□□	□□□□□
FR-WEB-001		/
FR-WEB-002		/terminal?deviceId=...
FR-WEB-003	deviceId	POST /api/device
FR-WEB-004	deviceId	PUT /api/device/{deviceId}
FR-WEB-005		DELETE /api/device/{deviceId}

4.2

□□	□□□□	□□□□
FR-MON-001		1
FR-MON-002	MQTT WebSocket	helmet/device/{deviceId}
FR-MON-003	X Y Z	50
FR-MON-004		0.0001
FR-MON-005	MQTT	

4.3

□□	□□□□	□□
FR-AV-001	helmetStatus SOS pushing	HTTP-FLV
FR-AV-002	WebRTC	{deviceId} _audio_web
FR-AV-003	SOS	
FR-AV-004	SOS pushing WebRTC	

5.

□□	□□□□□	□□□□
HTTP	GET http://localhost:11452/api/status	
HTTP	POST http://localhost:11452/api/device	deviceId label
HTTP	PUT http://localhost:11452/api/device/{deviceId}	
HTTP	DELETE http://localhost:11452/api/device/{deviceId}	
MQTT WebSocket	ws://localhost:8083/mqtt	
HTTP-FLV	http://localhost:8080/helmet/{deviceId} _video_terminal.live.flv	
WebRTC	http://localhost:8080/index/api/webrtc	{deviceId}_audio_web

6.

- MQTT 114514 IMEI
- MQTT 70 70
- GPS 30.593 114.305
-
- SOS
-
- API MQTT